

security. Despite these clear mandates, we have observed instances where even large system integration companies have found ways to circumvent these rigorous requirements, providing certifications with limited insight into the products they procure,” he points out.

It is no secret that China’s substantial global business volume, achieved through subsidies and economies of scale, allows it to offer products at competitive prices. Whether it’s toys or electronics, markets across the globe are flooded with products from the *World’s Factory*! Even in India, local partners often rebrand and resell these Chinese products. What raised alarm bells was when Subrahmanya said that most surveillance manufacturers in India provide limited declarations to the government. “The Indian government is mandated to purchase from the lowest bidder in public tenders, ultimately making way for cheaper products, due to their high volumes and subsidies, with weak firmware, whose source of origin remains ambiguous, into our national infrastructure,” he discloses.

Debugging the device

Any embedded device comprises three main components—a plastic casing, a motherboard containing the chip, and firmware, which holds the code. This firmware is primarily a Linux operating system code and is as complex as any other operating system code. Verma explains that cameras often have features such as TELNET (Teletype Network), FTP (File Transfer Protocol), DNS (Domain Name System) layers, and more, which are not typically necessary for their primary function. “These extra features can introduce additional vulnerabilities. To illustrate, it’s akin to leaving an email server running on a computer without proper security,” he says.

Subrahmanya emphasises the only conceivable way to uncover and under-



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stand these hidden functionalities is by delving into the firmware residing within the camera’s chip and acquiring the source code. “This extremely complex task necessitates a deep understanding of optoelectronic system programming. The firmware code can span millions of lines, making comprehensive testing an incredibly daunting endeavour,” he warns.

The enormous haystack of code conceals a tiny spark that could one day ignite a massive conflagration of security breaches. Subrahmanya delves deeper into the concern, stating, “Many internet-connected cameras, known as IP cameras, often come with a convenient feature called ‘zero config’ that simplifies installation by automatically sending video feeds to a default location when you connect the camera to an Ethernet cable and power it on. Typically, users configure these cameras with specific network addresses for surveillance purposes.”

“However, manufacturers possess additional ‘zero config’ settings

that aren’t always shared with partners deploying the cameras, which can be used to provide covert access to the camera by engaging a third party with knowledge of these hidden settings, granting unauthorised access to sensitive data streams,” he adds.

In the letter to Vaishnav, CAIT National President B C Bhartia and Secretary General Praveen Khandelwal expressed concerns that weak surveillance systems could easily transfer acquired data without competent data protection laws or monitoring mechanisms. The statement mentioned that IP cameras in CCTV networks and internet-operated digital video recording (DVR) for CCTV systems could be compromised at any given time.

So, does every camera pose a security risk? Verma replies in negative to this question. “The concept of ‘zero config’ relies on a protocol called UPnP (Universal Plug and Play). You simply plug in your camera and power it on, and it will automatically detect an IP address and start functioning. However, this convenience often comes at the